

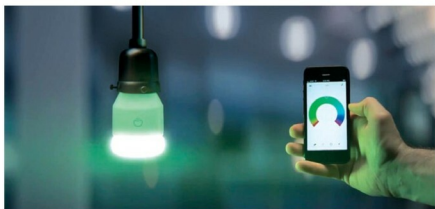
Technological Advancements in LEDs and ZigBee are Making Lighting Smart

Smart lighting is an interesting way to get started with automating your home, office or factory. It gives you immense flexibility and control, and provides great savings too, although there are some security concerns that need to be addressed. Let us take a look at how wireless technologies and advancements in LEDs are boosting smart lighting applications

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For most of us, light bulbs are basically 'turn on, turn off' devices. But today, smart lighting is changing the way we think about them. Light emitting diodes (LEDs) especially change a lot of things when incorporated into intelligent lighting systems. These use the inherent capabilities of LEDs (for example, instant dimming and on/off) to maximise energy savings and provide facility management with unprecedented control and flexibility over how lighting is used throughout the life cycle of lighting installation. It is possible to change the colour and control brightness levels of lights with an app on your Android or iOS mobile phone, and program them based on your lifestyle with the help of LEDs, sensors and wireless technologies.

Smart lighting systems integrate LED fixtures (with occupancy and daylight sensors) and wirelessly network them, while providing a software management and reporting interface. There is a wide variety of technologies that are being experimented for smart lighting products. The usage of wireless-connected devices has become extensive over the past decade or so. Technologies such as ZigBee, Bluetooth and Wi-Fi can be found in a variety of devices and their uses has become widespread—thanks to the boom in



The brightness levels of a light bulb can be adjusted using an app on your mobile phone

the smartphone industry. Lighting products, too, can benefit from this shift towards wireless-connected systems, that is the Internet of Things.

Emergence of ZigBee

"A widely-accepted standard is ZigBee, which has a specific profile for lighting applications called ZigBee light link. This helps enable a complete network for lighting and provides the ability to monitor and control its parameters," says Vikas Thawani, analog field applications engineer, Texas Instruments India. Given the number of mobile phones that have infiltrated the market, there is great interest from end-users to control lights using their mobile phones. Therefore, manufacturers are exploring wireless technologies, such as Bluetooth low-energy (LE) and Wi-Fi, on mobile phones. He adds, "We have actually seen a few custom-

ers work with Bluetooth LE, trying to interface their lights and provide an app for controlled lighting—turn on/off, dimming and occupancy based lighting. Wi-Fi is also being used to control the entire network of lighting."

Why ZigBee

ZigBee light link gives the lighting industry a global standard for interoperable and very easy-to-use consumer lighting and control products. "It allows consumers to gain wireless control over all their LED fixtures, light bulbs, timers, remotes and switches," says Jaiwin Ranjit, partner, Enlite Energy Solutions. He adds, "Products using this standard will let consumers change lighting remotely to reflect ambiance, task or season, all the while managing energy-use and making their homes greener."

Products built using this standard are as easy-to-use as a common dim-